Developmental Cell, Vol. 8, 467–477, April, 2005, Copyright ©2005 by Elsevier Inc. DOI 10.1016/j.devcel.2005.02.010

Temporal and Spatial Regulation of Phosphoinositide Signaling Mediates Cytokinesis

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Summary

Polarity is a prominent feature of both chemotaxis and cytokinesis. In chemotaxis, polarity is established by local accumulation of PI(3,4,5)P_3 at the cell's leading edge, achieved through temporal and spatial regulation of PI3 kinases and the tumor suppressor, PTEN. We find that as migrating *D. discoideum* cells round up to enter cytokinesis, PI(3,4,5)P_3 signaling is uniformly suppressed. Then, as the spindle and cell elongate, PI3 kinases and PTEN move to and function at the poles and furrow, respectively. Cell lines lacking both of these enzymatic activities fail to modulate PI(3,4,5)P_3 levels, are defective in cytokinesis, and cannot divide in suspension. The cells continue to grow and duplicate their nuclei, generating large multinucleate cells. Furrows that fail to ingress between nuclei are unable to stably accumulate myosin filaments or suppress actin-filled ruffles. We propose that phosphoinositide-linked circuits, similar to those that bring about asymmetry during cell migration, also regulate polarity in cytokinesis.

Introduction

Cytokinesis is the culminating event of the cell cycle that divides a parent cell into two daughter cells. In animals, the dividing cell usually rounds up, then elongates and begins to constrict at the midline. The membranes from opposite sides of the cell invaginate, then fuse, pinching off the newly formed progeny cells. In this dynamic process, cellular division is highly coordinated with mitotic segregation of the chromosomes and distribution of the cellular organelles. If the process is not properly regulated, components can be mislocalized or division can fail completely (Guertin et al., 2002; Robinson and Spudich, 2000; Schweitzer and D'Souza-Schorey, 2004; Wang et al., 2003). Cytokinesis is a potential therapeutic target for drugs, and its inhibition might prevent excessive cell proliferation as occurs in cancer (Miyamoto et al., 2003).

In cytokinesis, a coordinated network of dynamic events takes place at the furrow and poles of the dividing cell. A series of proteins, including myosin II, Rho GTPases, and cortexillin, associate with the furrow in multiple organisms, while others such as Rac E, dynacortin, coronin, and scar localize to the poles (Bi et al., 1998; De Lozanne and Spudich, 1987; Drechsel et al., 1997; Kishi et al., 1993; Kitayama et al., 1997; Lippincott and Li, 1998; Weber et al., 1999). Actin filaments play a key role at the poles as well as at the furrow. Newly polymerized actin and actin binding proteins associate with polar ruffles, while actin and myosin filaments accumulate in the furrow and contribute to the contractile ring that pinches the daughter cells apart (Fukui, 2000; Gerisch et al., 1999; Cao and Wang, 1996; O'Connell et al., 2001). Mutations or mislocalizations of these cytoskeletal proteins cause severe defects in cytokinesis. For example, in *Dictyostelium discoideum*, deletion of myosin II prevents cytokinesis when cells are grown in suspension. An important aspect of the cytokinesis machinery, however, is its resiliency. Many of these cytoskeletal proteins are individually dispensable for cytokinesis under specific conditions. For instance, when the myosin II null mutants are grown on surfaces, they are able to divide. In addition, *D. discoideum* cells can undergo "traction-mediated cytofission," in which multinucleated cells fragment and move apart.

Cells must acquire and maintain distinct cortical asymmetries for many critical cellular functions, including chemotaxis and cytokinesis. Using *D. discoideum* as a model system, the chemoattractant-induced mechanism that establishes asymmetry of migrating cells has been delineated: Receptor/G protein signaling directs PI3 kinases and PTEN to relocate to regions of the membrane exposed to higher and lower chemoattractant concentration, respectively (Devreotes and Janetopoulos, 2003; Funamoto et al., 2002; Iijima et al., 2002). PI(3,4,5)P_3 accumulates locally in the membrane at the front and aligns the cell along the gradient (Parent and Devreotes, 1999). The elevated phosphoinositide levels bias actin filament formation and pseudopodia extension to occur at the front. The decreased levels of PI(3,4,5)P_3 at the rear of the cell prevent extraneous projections in this region and help specify the back (Chen et al., 2003). In cells lacking PTEN, PI(3,4,5)P_3 and actin polymerization levels are coordinately elevated, the zone of pseudopodia formation is broadened, and the "polarity circuit" is compromised (Iijima and Devreotes, 2002). While the mechanisms of enzyme activation/inhibition have not been established, a similar local accumulation of PI(3,4,5)P_3 controls polarity in many cells, including neutrophils and fibroblasts (Haugh et al., 2000; Wang et al., 2002). In fact, loss of polarity and enhanced motility may contribute to the invasive and metastatic properties of mammalian cells lacking PTEN (Goberdhan and Wilson, 2003; Higuchi et al., 2001; Liliental et al., 2000).

While polarity during cytokinesis depends on internal cues rather than an external chemoattractant, the mechanisms for its establishment may be similar. Current thinking holds that during cytokinesis microtubules associated with the mitotic apparatus interact with and control the cell cortex and that the cortex, in turn, may affect the progress and orientation of the spindle (Cao and Wang, 1996; Rappaport, 1986; Rieder et al., 1997; Robinson and Spudich, 2004; Rosenblatt et al., 2004; Wheatley and Wang, 1996). Surprisingly, signaling events at the plasma membrane that may regulate the cy-

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toskeleton have received relatively little attention even though there have been reports that phosphoinositides and other phospholipids participate in cytokinesis (Brill et al., 2000; Iijima and Devreotes, 2002; Janssen and Schleicher, 2001). It seems natural that signaling events would play an important role in cytokinesis, as they do in cell migration. The bipolar shape of a cell during cytokinesis can be conceptualized as two oppositely polarized cells moving apart from each other, the poles being equivalent to the "fronts" of the two forming daughter cells, and the furrow corresponding to the "backs." Because of these morphological parallels between cell division and migration, we hypothesized that PI(3,4,5)P_3 might localize at the poles and influence the polarity of the actin cytoskeleton during cytokinesis. In fact, our results show that a "polarity circuit" involving the temporal and spatial regulation of PI(3,4,5)P_3 metabolism plays a key role in cell division.

Results

Signaling Components Function Locally during Cytokinesis

To test the hypothesis that the spatial distribution of PI(3,4,5)P_3 controls polarity during cell division, we examined distributions of a biosensor for this phosphoinositide, PH_{Crac}-GFP, in *D. discoideum* during cytokinesis. As shown in Figures 1A and 1B and Movie S1 (see the Supplemental Data available with this article online), prior to cytokinesis, PI(3,4,5)P_3 was localized to ruffles at the leading edges of migrating cells. As the cells rounded up at the onset of cytokinesis, PI(3,4,5)P_3 signaling was suppressed, and PH_{Crac}-GFP, no longer apparent on the membrane, was elevated in the cytosol (second panel in Figure 1B). Then, as cytokinesis progressed, the phosphoinositide concentrated on ruffles appearing at the poles of the dividing cell and was absent from the forming furrow (Figure 1C and Movie S2). Following division, PI(3,4,5)P_3 was again selectively localized on the anterior projections of the two daughter cells. In cells undergoing traction-mediated cytofission, there was a similar bipolar distribution of PI(3,4,5)P_3 on the membrane at the diverging ends of the cell (Figure S1; see the Supplemental Data available with this article online).

We next asked whether these local accumulations of PI(3,4,5)P_3 are controlled by reciprocal regulation of PI3 kinase (PI3K) and PTEN, as occurs in cell migration. In dividing cells, PI3K2-GFP was invariably associated with F-actin-based ruffles occurring at the poles. In contrast, PTEN-GFP was associated with the membrane in the central region of the cell that was devoid of ruffles. These localizations were similar in cells undergoing traction-mediated cytofission (data not shown). The distributions were maintained as the process of division progressed, and, as the cells separated, PI3K2-GFP was sharply localized at the anterior of the two daughters, while PTEN-GFP was clustered at the rear (Figure 1D; Movies S3 and S4). Furthermore, as shown in Figure 1E, when cells rounded up to enter cytokinesis, PTEN-GFP intensely associated with the entire membrane. This distribution is consistent with the suppression of PI(3,4,5)P_3 signaling observed during this phase (Figures 1A and 1B). As the spindle elongated, PTEN dissociated from the poles and concentrated at the middle and along the lateral edges. These movements of PTEN mirror those of myosin II (see Figure 5B below). Importantly, as shown in Figure 1E, the actin binding protein Coronin-GFP redistributed in coordination with PI(3,4,5)P_3, as occurs in migration (Chen et al., 2003).

We postulated that the reciprocal distribution of PI3K and PTEN would create an asymmetric sensitivity to uniform chemoattractant as it does in polarized migrating cells (Jin et al., 2000). We applied a uniform stimulus of folic acid, a chemoattractant active in growing cells, to cells in the midst of cytokinesis and imaged the redistribution of PH_{Crac}-GFP to monitor changes in PI(3,4,5)P_3 levels. The stimulus elicited a burst of PI(3,4,5)P_3 at the ruffled poles of the dividing cell with little or no accumulation in the smooth central region (Figure 1F). Cytokinesis proceeded throughout the chemoattractant-induced response, which lasted 30–60 s. This spatially restricted sensitivity could be observed at the earliest stages of spindle elongation. The experiment shown in Figure 1F is included as Movie S5. Similar results were obtained with cells stimulated with chemoattractant at earlier stages of cytokinesis.

Perturbation of the Spatial Distribution of PI(3,4,5)P_3 Inhibits Cytokinesis

To test whether the localized distributions of PI(3,4,5)P_3 played a role in cytokinesis, we examined the growth of existing mutants with altered phosphoinositide levels. As shown in Figure 2A, deletion of PI3K1 and PI3K2, which lower PI(3,4,5)P_3 levels, or deletion of PTEN, which prolongs chemoattractant-induced changes in PI(3,4,5)P_3, each increased the doubling time. Similarly, PI3K inhibitors interfered with cytokinesis (see below). The effects of inactivation of PTEN were stronger (Huang et al., 2003; Iijima and Devreotes, 2002), suggesting that preventing the degradation of PI(3,4,5)P_3 is more deleterious than lowering levels. However, even with inhibitors or in the *pi3k*$^{1\text{-}2-}$ or *pten*$^{-}$, cells there still is modulation of PI(3,4,5)P_3. This is because each activity is independently regulated in a reciprocal manner (Chen et al., 2003; Ma et al., 2004). In order to further impair regulation, we perturbed both PI3K and PTEN activities. Starting with the cell line carrying deletions in PI3K1 and PI3K2 (*pi3k*$^{1\text{-}2-}$ cells), we also deleted PTEN, generating a new line lacking all three enzymes (*pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells). The Southern blot in Figure S1 demonstrates that the PTEN locus was disrupted as predicted. As anticipated, the deletion of both activities generated a cell line that lacked temporal and spatial regulation of PI(3,4,5)P_3 (see below).

As shown in Figure 2A, these cells were unable to divide in suspension, and we sought the origin of the defect. As the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells were shaken for several days, they became increasingly large. The fold increases in protein mass during overnight culture were essentially the same for the wild-type and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells (4.8 and 4.9, respectively). As shown in Figure 2, when examined by Nomarski DIC optics after 3 days, the diameters of the largest cells ranged up to 120 μm, whereas the typical wild-type cell was about 15 μm. Although the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells failed to increase in

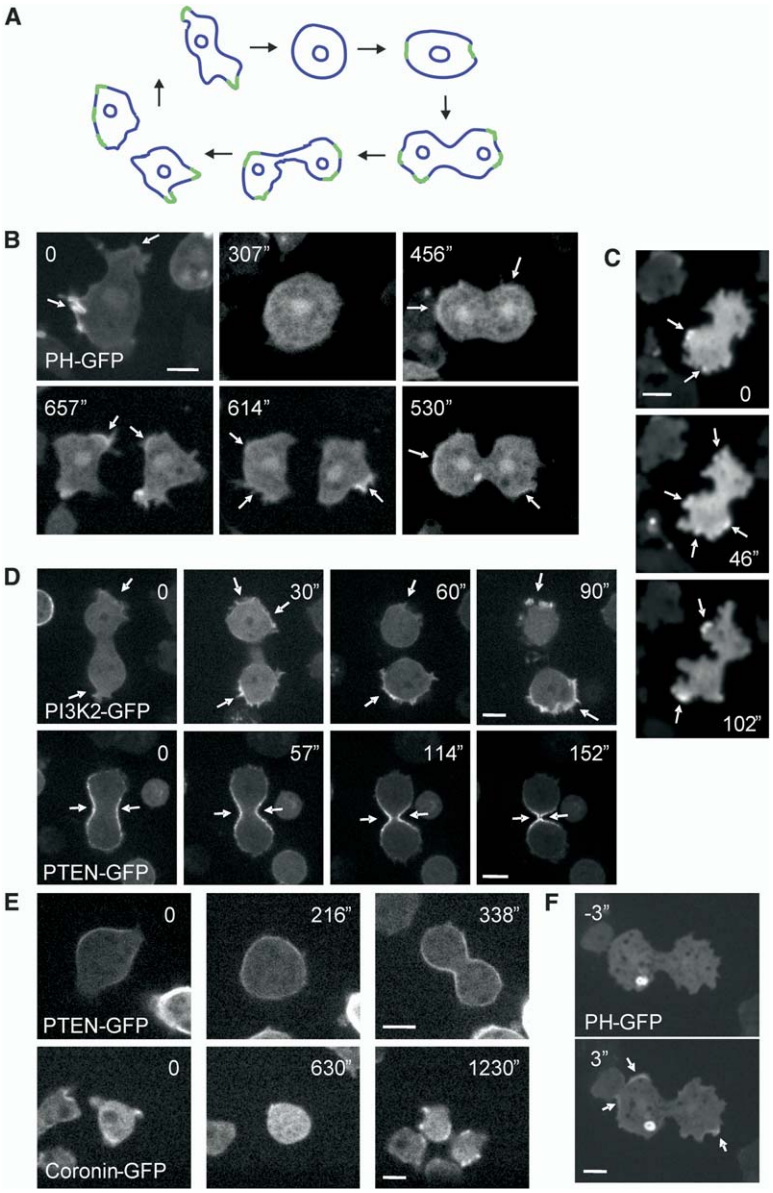

Figure 1. Localization of PI(3,4,5)P_3, PI3Ks, and PTEN during Cytokinesis

Wild-type cells expressing PH_{Crac}-GFP, PI3K2-GFP, or PTEN-GFP were examined, and cell division events were recorded.

(A) Cartoon of the distribution of PI(3,4,5)P_3 at stages during cytokinesis. Green patches on the membrane represent local accumulations of PI(3,4,5)P_3; nuclei within cells are also depicted. See the text for explanation.

(B) Single cell expressing PH_{Crac}-GFP imaged before and during cytokinesis. The scale bar is 5 μm.

(C) Time sequence of PH_{Crac}-GFP during cytokinesis, imaged by TIRFM. Arrows in (B) and (C) point to local accumulations of PH_{Crac}-GFP bound to the membrane.

(D) Distributions of PI3K2-GFP and PTEN-GFP during cytokinesis.

(E) PTEN-GFP (top) and Coronin-GFP (bottom) were imaged at the onset of cytokinesis. The bars in (C)—(E) are 8 μm. The times indicate seconds after the start of imaging.

(F) Cells undergoing cytokinesis, expressing PH_{Crac}-GFP, were imaged 3 s before and after addition of 10 μM folic acid. The arrows show the appearance of PI(3,4,5)P_3 at the poles. The scale bar is 5 μm.

number, the nuclei continued to divide. The rate of nuclear division was essentially identical in the *pi3k*$^{1\text{-}2-}$ *pten*$^-$ and wild-type cells (Figure 2B), indicating that the defect was specific for cytokinesis and that the other stages of the cell cycle were apparently normal. At steady state, less than 15% of the *pi3k*$^{1\text{-}2-}$*pten*$^-$ cells were mononucleated, whereas over 75% had six or more nuclei (Figure 2C). The *pi3k*$^{1\text{-}2-}$ and *pten*$^-$ cultures also contained multinucleated cells. About 20% had up to five nuclei, and about 3% contained six to ten. In contrast, nearly 75% of the wild-type cells were mononucleated, and none had more than five nuclei. We also determined the distribution of nuclei in cultures of *gβ*$^-$ cells, which lack the unique Gβ subunit of heterotrimeric G proteins found in *D. discoideum*. Interestingly, these populations contained over 95% mononucleated cells, suggesting that G protein signaling may interfere with cytokinesis (data not shown).

We also examined the cells grown on plastic surfaces, which as described above, is a less stringent condition than suspension growth (Uyeda et al., 2000). Cells lacking important components for cytokinesis such as myosin II, cortexillin, or Rac E, for example, which cannot grow in suspension, are often able to divide on surfaces. Likewise, the *pi3k*$^{1\text{-}2-}$*pten*$^-$ cells were able to grow under this condition. On surfaces, the *pi3k*$^{1\text{-}2-}$, *pten*$^-$, and *pi3k*$^{1\text{-}2-}$*pten* cells still contained abnormally sized cells (Figures 2D–2F and data not shown; Movies S6 and S11 shows a *pten*$^-$ cell and a *pi3k*$^{1\text{-}2-}$*pten*$^-$ cell failing cytokinesis). Fields of the *pi3k*$^{1\text{-}2-}$*pten*$^-$ cultures examined by phase microscopy contained cells with diameters up to 120 μm. Expression of PTEN-GFP or human PTEN-YFP (hPTEN-YFP) in the *pi3k*$^{1\text{-}2-}$*pten*$^-$ cells completely reversed these defects (Figure 2D). $PTEN_{16–515}$-YFP and $PTEN_{C124S}$-YFP did not eliminate the large-cell phenotype (data not

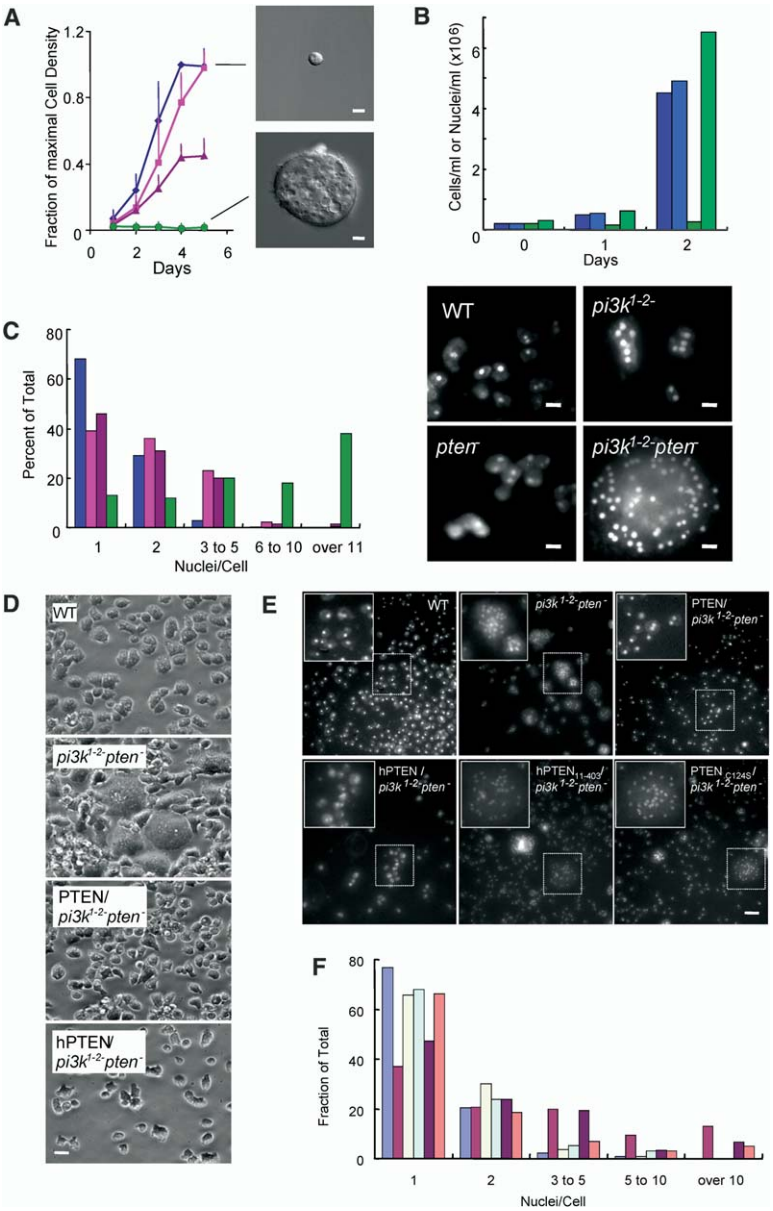

Figure 2. Analysis of Growth in Wild-Type and Mutant Cell Lines

(A) Curves show growth of shaking cultures of wild-type (blue diamonds), *pi3k*$^{1\text{-}2-}$ (pink squares), *pten*$^{-}$ (purple triangles), and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (green circles) cells. All values in an experiment were normalized to the wild-type value at day 4, which was typically 1–2 × 10^7 cells/ml. Experiments were repeated four times, and averages and standard deviations are shown. Nomarski DIC images show typical wild-type (upper) and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (lower) cells after 3 days of shaking culture. The scale bar is 15 μm.

(B) Wild-type and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells were grown in shaking conditions. Cells and DAPI-stained nuclei were counted. Wild-type cells (dark blue), wild-type nuclei (light blue), *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells (dark green), *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ nuclei (light green).

(C) Wild-type (dark blue), *pi3k*$^{1\text{-}2-}$ (pink), *pten*$^{-}$ (purple), and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (green) cultures where shaken for 3 days, plated, and stained with DAPI. Random fields were photographed and fractions of cells containing the designated number of nuclei were determined. Averages of three experiments comprising over 200 cells for each line are shown. Selected epifluorescence images of mononucleated wild-type and multinucleated *pi3k*$^{1\text{-}2-}$, *pten*$^{-}$, and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells are shown. The scale bars are 10 μm.

(D) Cells adhering to a coverslip were grown for 3 days and were imaged by 400× phase contrast microscopy. Typical fields of wild-type, *pi3k*$^{1\text{-}2-}$*pten*$^{-}$, as well as PTEN-GFP or hPTEN-YFP expressed in *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells, are shown. The scale bar is 10 μm.

(E) Random fields of DAPI-stained cultures of the indicated cell lines grown on surfaces. Insets show enlargements of selected portions of each field. The scale bar is 30 μm.

(F) Cultures of wild-type (dark blue), *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (maroon), PTEN/*pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (yellow), hPTEN/*pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (light blue), hPTEN$_{10\text{-}403}$/*pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (purple), and PTEN$_{C124S}$/*pi3k*$^{1\text{-}2-}$*pten*$^{-}$ (salmon) cells were grown on surfaces and stained with DAPI. Fractions of cells with designated numbers of nuclei are shown. Averages of two experiments comprising over 200 cells for each line are shown.

shown; see below). We have previously shown that PTEN$_{16\text{–}515}$-YFP, which lacks the N-terminal PI(4,5)P_2 binding motif, fails to interact with the membrane or rescue the chemotaxis defect of the *pten*$^{-}$ cells. PTEN$_{C124S}$-YFP displays low catalytic activity, but it does interact with the membrane and partially rescues the aggregation defect (Iijima et al., 2004). The human counterparts of these mutants, hPTEN$_{11\text{–}403}$-YFP, which does not interact with the membrane (unpublished data), and catalytically inactive hPTEN$_{C124S}$-YFP were ineffective in reversing the large-cell phenotype.

DAPI staining showed that *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells grown on surfaces were a mixed population containing mononucleated cells as well as multinucleated cells containing up to 100 nuclei (Figure 2E). The mononucleated cells derived from traction-mediated cytofission as well as cytokinesis (data not shown; see below). Expression of PTEN-GFP or hPTEN-YFP yielded primarily mononucleated cells, whereas hPTEN$_{11\text{–}403}$-YFP did not eliminate the multinucleated cells. PTEN$_{C124S}$-YFP shifted the distribution toward mononucleated cells but did not eliminate the multinucleated phenotype. Quantification of the distributions of nuclei in the wild-type and *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells and the effects of expression of PTEN-GFP, hPTEN-YFP, hPTEN$_{11\text{–}403}$-YFP,and PTEN$_{C124S}$-YFP in the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells are shown in Figure 2F.

We also perturbed PI(3,4,5)P_3 levels in wild-type and *pten*$^{-}$ cells by addition of the PI3K inhibitors LY294003 and wortmannin. We searched for cells entering or in the midst of the cytokinesis process, added the drugs, and observed the immediate effects. The need for imaging restricted our studies to cells on surfaces on which cytokinesis defects are typically less penetrant (see Figure 2). Nevertheless, about 25% of the events

Table 1. Defective Cytokinesis Induced by PI3K Inhibitors

	Total	Fail	Delayed	Defective
Wild-Type				
Control	30	0	0	0%
DMSO	16	0	1	6%
LY294003	28	5	4	32%
pten⁻				
Control	37	1	0	3%
DMSO	21	0	2	9%
LY294003	31	3	4	23%
Wortmannin	20	2	3	25%

were perturbed by either drug treatment (Table 1). The cells either failed to divide, or the process was extremely delayed. This failure rate on surfaces is comparable to that observed for cells lacking myosin II (De Lozanne and Spudich, 1987; Knecht and Loomis, 1987). Examples of the failed attempts at cytokinesis in wild-type and *pten*⁻ cells are included as Movies S7 and S9.

Cells Lacking Dual PI3K/PTEN Regulation Are Unable to Respond to External Cues

We reasoned that our genetic perturbations eliminated modulation of PI(3,4,5)P_3 levels by internal cues operating during cytokinesis and that this lack of regulation would also be reflected in the temporal response to chemoattractant. We examined the regulation of PI(3,4,5)P_3 levels during stimulation with the growth-stage chemoattractant, folic acid. As shown in Figure 3, wild-type cells responded to a uniform increase in folic acid with a transient production of PI(3,4,5)P_3. While no PI(3,4,5)P_3 was observed in the *pi3k*$^{1\text{-}2-}$ cells in this assay, more sensitive assays have shown that PI(3,4,5)P_3 levels are low but can still be modulated in differentiated *pi3k*$^{1\text{-}2-}$ cells stimulated with cAMP (Huang et al., 2003; Meili et al., 2000). As previously shown in the *pten*⁻ cells, PI(3,4,5)P_3 levels were elevated in the absence of stimulation, and the response to folic acid was larger and more prolonged than in the wild-type cells, as has been observed previously for cAMP stimuli (Iijima and Devreotes, 2002). As anticipated, in the *pi3k*$^{1\text{-}2-}$*pten*⁻ cells, basal levels were also elevated, but exposure to folic acid had no further effect. Differentiated *pi3k*$^{1\text{-}2-}$*pten*⁻ cells also had high uniform levels of membrane PI(3,4,5)P_3 that were insensitive to cAMP stimulation (data not shown). To assess the integrity of the signaling system, we also stimulated *pi3k*$^{1\text{-}2-}$*pten*⁻ cells expressing the isolated membrane targeting domain of PI3K2 (N600-GFP) and found that it translocated normally to the membrane.

We quantified these responses in resting cells and at 5 and 30 s after stimulus addition by obtaining the ratio of signal within a small region on the membrane and a region of the cytosol. The changes in the ratios as a function of time reflected the changes seen in the images. In the wild-type and *pi3k*$^{1\text{-}2-}$ cells, the "basal" ratios were low, as is apparent from the images. In the *pten*⁻ and *pi3k*$^{1\text{-}2-}$*pten*⁻ cells, the prestimulus ratios were significantly elevated and were similar to each other. The higher basal PI(3,4,5)P_3 levels correlated with a higher level of membrane protrusions including an unusual number of PI(3,4,5)P_3-labeled filopodia. Images of the wild-type and *pi3k*$^{1\text{-}2-}$*pten*⁻ cells expressing

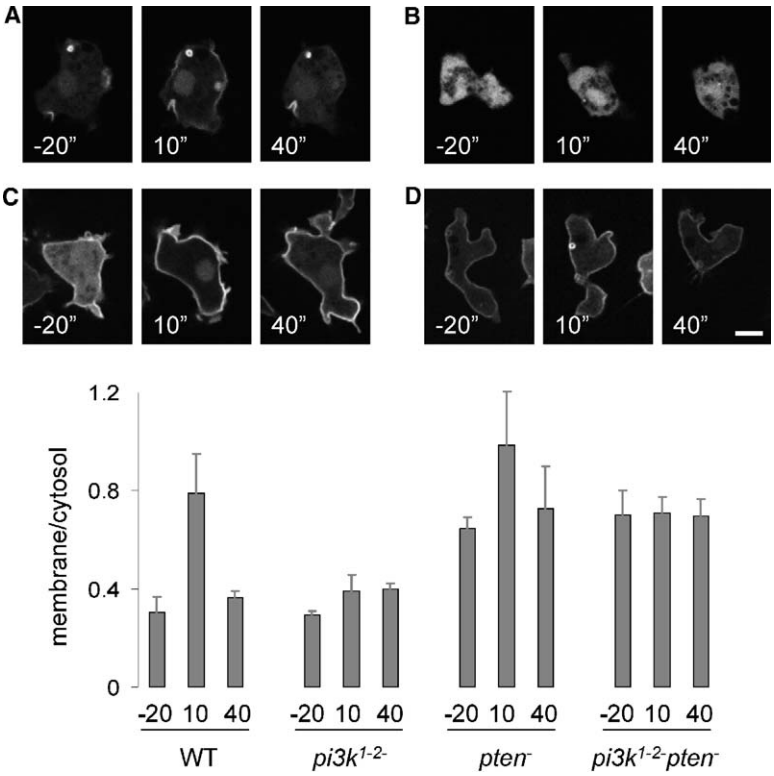

Figure 3. PI(3,4,5)P_3 Accumulation in Wild-Type and Mutant Cells during Folic Acid Stimulation

(A–D) (A) Wild-type, (B) *pi3k*$^{1\text{-}2-}$, (C) *pten*⁻, and (D) *pi3k*$^{1\text{-}2-}$*pten*⁻ cells expressing PH_{Crac}-GFP were stimulated with a uniform concentration (10 μm) of folic acid. The scale bar is 5 μm. Images and plots of the membrane:cytosol ratio of each cell line 20 s before the addition of folic acid, during the peak response at 10 s, and after 40 s of stimulation are shown. A box centered on the membrane also encompasses a portion of the cytosol. We therefore estimated the membrane signal by subtracting 50% of the cytosolic value from the value in the box centered on the membrane. Each time point indicates the average and standard error of six cells from two independent experiments.

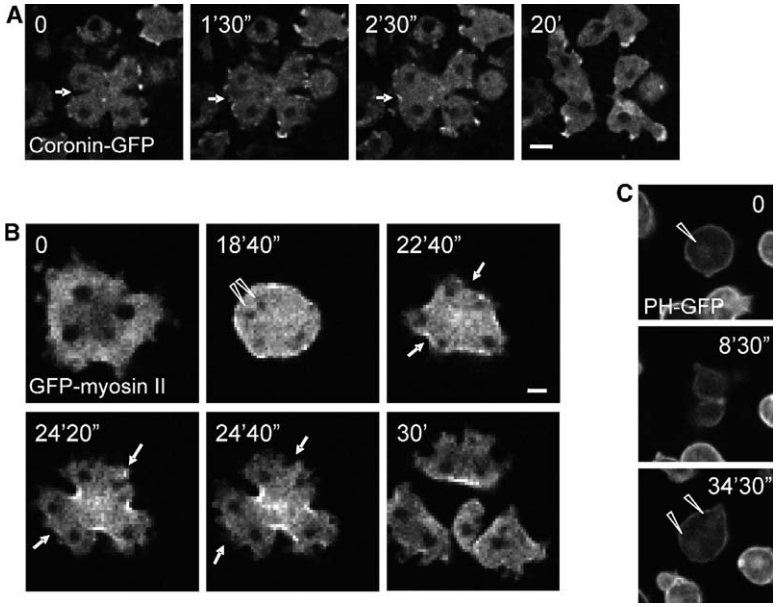

Figure 4. Coronin-GFP, GFP-Myosin II, and PH_{Crac}-GFP Distribution in *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ Cells during Cytokinesis

(A) Cell expressing Coronin-GFP. The furrow (see arrow) fails to constrict.

(B) Nuclei in multinucleated *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells expressing GFP-myosin divide normally. Some furrows succeed in separating the nuclei, while others fail. The arrowheads in the second frame show a dividing nucleus. The arrows in the subsequent frames show furrows that fail to ingress and begin ruffling.

(C) Distribution of $PI(3,4,5)P_3$ during failed cytokinesis. The arrowheads point to mononucleate and binucleate cells before and after mitosis. Images were taken at the indicated seconds after the start of imaging. The scale bars are 5 μm.

PH_{Crac}-GFP and examined by total internal reflection fluorescence microscopy (TIRFM) are included as Figure S2. The effects of these mutations on $PI(3,4,5)P_3$ levels are consistent with direct measurements of PI3K and PTEN activities (see Discussion).

Asymmetric $PI(3,4,5)P_3$ Distribution Influences Localization of Cytoskeletal Proteins

To further investigate the defects in *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells, we imaged the distribution of Coronin-GFP, an actin binding protein that labels actin-based projections at the front of migrating cells and at the poles of dividing cells (Gerisch et al., 1995); GFP-myosin II, which accumulates in the furrow of dividing cells and plays a major role in cytokinesis (De Lozanne and Spudich, 1987; Knecht and Loomis, 1987); as well as PH_{Crac}-GFP. Figure 4 illustrates striking examples in which furrow formation fails between some dividing nuclei and succeeds between others. Coronin-GFP was normally associated with ruffles along the cell cortex in migrating *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells. During cytokinesis, Coronin-GFP is suppressed in the midzone and furrows. However, in furrows that fail, Coronin-labeled ruffling reappears as the constriction is released (Figure 4A and Movie S9). In contrast, GFP-myosin II concentrated in forming furrows, but then it suddenly dispersed as the furrow was replaced with cortical ruffles on the forming binucleated cell (Figure 4B). Thus, the failure of Coronin-GFP suppression and GFP-myosin II concentration in the furrow both correlated strongly with aborted cytokinesis. For cells on surfaces, about 40% of the failed furrows aborted in this manner (n = 37). This experiment is included as Movie S10. We never observed a furrow that failed to properly localize the cytoskeletal proteins and succeeded to divide. On the other hand, in 60% of failed furrows, cytoskeletal proteins localized properly, but then the cell failed to sever the narrow bridge connecting the two daughters. Eventually, the bridge increased its diameter and the two cells flowed back together (Movie S10). This phenotype resembles the kinesin Kif12 mutants recently described (Lakshmikanth et al., 2004). In contrast to the cytoskeletal proteins, $PI(3,4,5)P_3$ remained uniformly distributed along the cell perimeter (Figure 4C), whether the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells succeeded or failed to divide. Movie S11 shows an example of a cytokinesis failure in *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells expressing PH_{Crac}-GFP.

Localization of PI3K and PTEN Depends on Mitosis

To test the role of the spindle microtubules in the regulation of $PI(3,4,5)P_3$ during cytokinesis, we treated cells expressing PH_{Crac}-GFP or PTEN-GFP with the mitotic inhibitor nocodazole. As previously shown, the inhibitor had little effect on motility but delayed cells at mitosis (Kitanishi-Yumura and Fukui, 1987; Graf et al., 2003). $PI(3,4,5)P_3$ accumulation on ruffles during random migration was normal, suggesting that the regulation of PI3K and PTEN was unperturbed. As cells rounded-up to enter cytokinesis and the ruffles subsided, $PI(3,4,5)P_3$ disappeared, PH_{Crac}-GFP accumulated in the cytosol, and PTEN-GFP became uniformly associated with the membrane. These early events were indistinguishable in treated or untreated cells (compare Figure 1 and Figure 5). However, whereas in untreated cells, this quiescent stage lasted less than 1 min, in nocodazole, it persisted. After an hour, about 10% of the treated cells, compared with only 0.5% of untreated cells, displayed this profile. When the nocodazole was washed out, the arrested cells resumed cytokinesis and $PI(3,4,5)P_3$, PI3K-GFP, and PTEN-GFP localized appropriately (Figure 5B and data not shown).

Discussion

Our results show that an elegant regulatory circuit involving phosphoinositide signaling, similar to the sys-

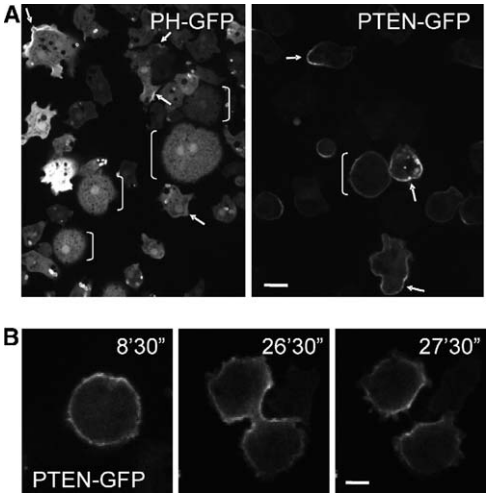

Figure 5. Effects of Nocodazole Treatment on Wild-Type Cells Expressing PH_{CRAC}-GFP or PTEN-GFP

(A) The panel on the left shows interphase cells with PI(3,4,5)P_3-labeled membrane protrusions (arrows), while other cells (brackets) stall at the G2/M transition and are quiescent. The right panel shows the localization of PTEN-GFP at the rear of interphase cells (arrows) and the uniform accumulation of PTEN-GFP at the membrane of a cell stalled at entry to mitosis (bracket). The scale bar is 5 μm.

(B) PTEN-GFP cells were treated for 3 hr with nocodazole, washed, and observed for the times indicated. Cells underwent cytokinesis, and PTEN-GFP localized normally.

tem that sets direction in chemotaxis, controls polarity in cytokinesis (Figure 6). In chemotaxis, receptors and G proteins control the spatial accumulation of PI(3,4,5)P_3, whereas internal cues, presumably from the mitotic apparatus, drive cytokinesis (Devreotes and Zigmond, 1988; Guertin et al., 2002). In chemotaxis, PI(3,4,5)P_3 accumulates on the leading edge, and perturbations of the PI3K/PTEN polarity circuit adversely affect migration. In cytokinesis, we find strikingly parallel redistributions of PI3K and PTEN. As cells round at the onset, PTEN moves uniformly to the membrane. Then, as the cell elongates, PI3K and PTEN associate with the membrane at the poles and furrow, respectively (Figure 6). Our disruption of both PI3K and PTEN activities generated cells in which PI(3,4,5)P_3 could not be modulated. This led to a specific failure in cytokinesis, resulting in large, multinucleated cells. Expression of active PTEN from *D. discoideum* or human reversed the defect. This is the first report that regulation of PI(3,4,5)P_3 metabolism plays a role in cytokinesis. However, considering the general role of PI(3,4,5)P_3 in migration, it is likely that the mechanism we describe here for cytokinesis will be found in other organisms (Hannigan et al., 2004; Wang et al., 2002). A recently reported association of PTEN with the septa of *S. pombe* may be related (Mitra et al., 2004).

Based on studies of migrating cells, we can speculate on the role of PI(3,4,5)P_3 in cytokinesis. In a migrating cell, there is a steep internal gradient of PI(3,4,5)P_3 that promotes actin-filled projections selectively at the front. Actin filaments and myosin accumulate in the area devoid of PI(3,4,5)P_3 at the rear and provide contraction. Similarly, the regulation of PI(3,4,5)P_3 during cytokinesis may bias the activity of the cytoskeleton. At the onset of cytokinesis, PTEN moves to the membrane, and PI(3,4,5)P_3 signaling is uniformly suppressed, leading to a loss of membrane ruffling and actin polymerization. This may allow the cell to erase polarity established during cell migration. As the cell elongates, the poles gradually become active zones of PI(3,4,5)P_3-labeled, actin-filled projections, whereas the midzone and forming furrow are devoid of the phosphoinositide, like the rear of a migrating cell. This absence of PI(3,4,5)P_3 from the furrow inhibits actin-based projections in this region and may permit the acto-myosin-based contractions to occur. This segregation of cytoskeletal activities is consistent with previ-

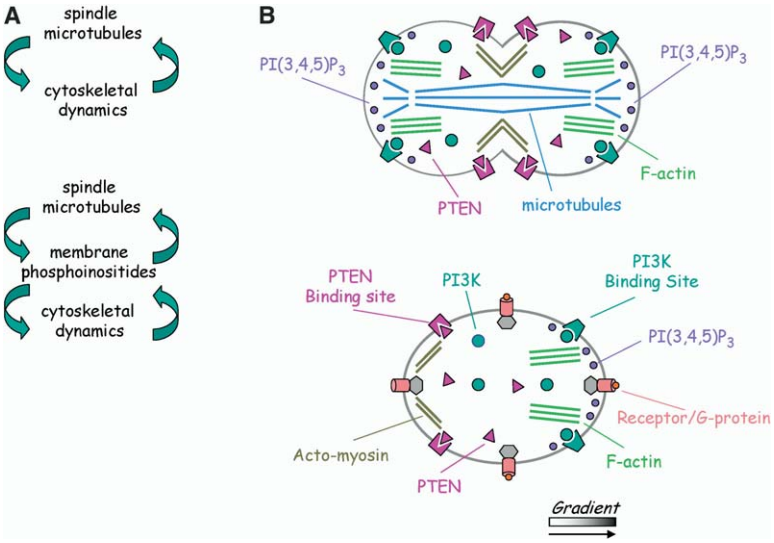

Figure 6. Roles of the "Phosphoinositide-Linked Polarity Circuit" in Cytokinesis and Chemotaxis

(A) Left, top: current models envision a reciprocal interaction between the spindle microtubules and cytoskeletal dynamics that work together to form the furrow and drive cytokinesis. Evidence for and against the roles of the depicted interactions in various cell types have been extensively reviewed (Guertin et al., 2002; Oegema and Mitchison, 1997). Left, bottom: The scheme proposed here introduces the additional participation of membrane phosphoinositides. Phosphoinositides modulate cytoskeletal dynamics, which in turn regulate cytokinesis.

(B) Right, top: cartoon of a dividing cell showing the distribution of components during cytokinesis. Internal cytokinesis cues from the spindle microtubules affect membrane and cytoskeletal dynamics. Binding sites for PI3K and PTEN move to the poles and furrow, respectively. PI(3,4,5)P_3 synthesis and actin polymerization occurs selectively at the poles, while actomyosin concentrates in the furrow. Right, bottom: distribution of components during chemotaxis. Uniformly distributed receptors and G proteins elicit the recruitment of PI3K and PTEN to the front and back of the cell, respectively. PI(3,4,5)P3 accumulation at the front promotes actin polymerization.

ous studies of normal rat kidney cells in which local application of actin polymerization inhibitors to the poles prevents cytokinesis, whereas treatment of the furrow accelerates the process (O'Connell et al., 2001). It is also possible that concurrent elevation of PI(4,5)P_2 in the furrow may contribute to the polarization of the cell (Brill et al., 2000). In fact, PTEN contains a PI(4,5)P_2 binding motif that could possibly direct it to the furrow of a dividing cell or the back of a migrating cell. However, we have no direct evidence that PTEN actually binds to PI(4,5)P_2 or that this phosphoinositide is elevated in these regions.

The inability to regulate PI(3,4,5)P_3 and influence the activity of the cytoskeleton in the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells likely contributes to the observed failure at multiple stages of cytokinesis. First, it would be more difficult for these cells to reset polarity at the onset of cytokinesis. Second, the uniform distribution of PI(3,4,5)P_3 would be unable to reinforce the segregation of cytoskeletal activities during furrow formation. In fact, in the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells imaged on surfaces, nascent furrows lacking Coronin-GFP and with accumulated GFP-myosin II initially formed, but then, the myosin was often lost, and the furrow was replaced with Coronin-labeled, actin-based ruffles. While we could not image under shaking conditions in which the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells always fail cytokinesis, we presume that the role of PI(3,4,5)P_3 to properly localize and stabilize the cytoskeletal proteins is critical. The fact that even the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells were able to occasionally divide when grown on surfaces is not surprising since cells lacking other critical components that mediate cytokinesis, including myosin II, cortexillin, and Rac E, display the same phenotype. While the rate of the cell cycle was unaffected in the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells, it is likely that they are defective in additional processes, such as macropinocytosis and phagocytosis, which require PI(3,4,5)P_3 regulation of the cytoskeleton.

Our disruption of PTEN in a cell already lacking PI3K1 and PI3K2 removed most of the temporal and spatial regulation of PI(3,4,5)P_3, but it did not eliminate the basal levels. It has been shown directly that PI3K1 and PI3K2 contribute most of the regulated, but little of the basal, PI3K activity (Huang et al., 2003). An additional class I and a class III PI3K (PIK3 and PIK5, respectively) have been previously reported (Meili et al., 1999; Zhou et al., 1995). Our observation prompted us to reexamine the database for additional PI3Ks. We found that the database includes four additional enzymes. An analysis of the PIKs, together with suggested nomenclature for the new enzymes, is included as Figure S4. Presumably, some of these PI3Ks contribute the basal activity. A constant source of PI(3,4,5)P_3 synthesis could explain the observations in Figure 3 as follows: in wild-type and *pi3k*$^{1\text{-}2-}$ cells, the basal levels of PI(3,4,5)P_3 are held in check by PTEN, and the transient dissociation of PTEN from the membrane contributes to the modulation of PI(3,4,5)P_3. When PTEN is deleted, basal levels increase, but modulation of PI(3,4,5)P_3 is still supplied by activation of PI3K1 and PI3K2. But in *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells, only the unchecked constant basal levels remain.

Our experiments with nocodazole suggest that events depending on the spindle microtubules control the spatially localized activities of PI3K and PTEN. As cells round-up at the onset of cytokinesis, PTEN and myosin II associate uniformly with the membrane, and PI(3,4,5)P_3 signaling is suppressed. It is also known that the microtubule network is disassembled at this stage (Neujahr et al., 1998). Cells treated with nocodazole are delayed at this stage because the spindle cannot be easily constructed. Under these conditions, the polarization of PI3K and PTEN, PI(3,4,5)P_3 signaling, and membrane ruffling are concomitantly delayed. This suggests the existence of a checkpoint that prevents further progression into cytokinesis if the spindle fails to assemble. We speculate that the elongation of the spindle leads to the localization of PI3K and PTEN to the membrane at the poles and furrow, respectively. Perhaps specific proteins link ends of the astral microtubules to the actin network (Canman and Bement, 1997; Goode et al., 2000; Hestermann et al., 2002). For directed migration, the upstream control derives from G protein-coupled receptor (GPCR) activation, whereas cytokinesis clearly does not require G protein signaling and is driven by intrinsic cues (Devreotes and Zigmond, 1988; Guertin et al., 2002). Yet, in both cases, the signals somehow trigger local membrane binding of PI3Ks and PTEN. Since the two processes share these elements, it is possible that the pathways may interact at the level of membrane binding sites for PI3K and PTEN or at targets for PI(3,4,5)P_3. We have noted that shaking cultures of wild-type cells contain about 25% multinucleated cells and that these multinucleated cells are absent from *gβ*$^{-}$ cultures. This may point to an antagonism between the GPCR signaling and cytokinesis pathways. Other recent reports have suggested undefined positive or negative roles for G protein signaling in cytokinesis (Fuse et al., 2003; Grill et al., 2003; Manning, 2003).

It is well known that defects in regulation of PI(3,4,5)P_3 are a major determinant in cancer (Phillips et al., 1998; Vivanco and Sawyers, 2002). One common mechanism of elevating PI(3,4,5)P_3 levels is by inactivation of the tumor suppressor PTEN. Mutations in PTEN are found in a large fraction of human cancers (Vazquez and Sellers, 2000). Recent evidence also has shown that mutations in PIK3CA, which increase kinase activity, are found in many colorectal tumors (Samuels et al., 2004). The proposed role of elevated PI(3,4,5)P_3 is to promote cell growth and survival. Other experiments show that activation of cell motility occurs with elevated PI(3,4,5)P_3 levels and may play a role in tumor progression (Chung et al., 2004; Wu et al., 2003). Our findings presented here suggest yet another potential role for the phosphoinositide in cancer. Although elevation of PI(3,4,5)P_3 levels does make cells grow faster, perturbation of its proper regulation may also promote disease by interfering with proper cell division. Moreover, defects in cytokinesis may also lead to defects in chromosome integrity, leading to aneuploidy, commonly found in many cancers. Many cancers appear to progress most rapidly once cells become aneuploid (Bose et al., 2002). In this regard, we have noted that a small fraction of the *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ cells have extremely enlarged nuclei and unbalanced nuclear divisions. Thus, disruption of a PI(3,4,5)P_3 polarity circuit regulat-

ing cytokinesis like that described here may significantly contribute to tumor progression.

Experimental Procedures

Cell Growth and Development

All cell lines were cultured axenically in HL5 medium (Huang et al., 2003) at 22°C. Transformants were maintained in 20 μg/ml G418, 30 μg/ml Hygromycin, or both as required. Cells stimulated with folic acid were grown on nutrient SM agar plates with *Klebsiella aerogenes* (Janetopoulos et al., 2004).

Construction of *pi3k*$^{1\text{-}2-}$*pten*$^{-}$ Cells

To construct the PTEN disruption vector, the Hygromycin resistance cassette was inserted into the BamH1 site in the PTEN coding region and cloned into the TA cloning vector pCR2.1 (Invitrogen). The linearized disruption vector (10 μg) was introduced into growth-phase *pi3k*$^{1\text{-}2-}$ cells (Funamoto et al., 2001).

Vector Constructs

HPTEN-YFP and deletion mutants fused to YFP were amplified by PCR from pSG5L-HA-PTEN;WT (Vazquez et al., 2000) and cloned into pJK1.

Microscopy

Wide-field epifluorescence microscopy images of DAPI-stained and living cells were obtained by using an inverted Zeiss Axiovert 100 TV with a 40× 1.3 N.A. Achrostigmat lens and were collected by using a Roper Photometrics CoolSNAP camera. Confocal images were obtained by using a Perkin Elmer UltraView system on a Nikon inverted microscope with a 63× 1.4 N.A. Apochromat lens (Figures 1B, 2A–2C, 4A–4D, 5A–5C, and 6). TIRFM images were acquired on an Olympus IX71 objective-based TIRF configuration with a 60× 1.45 N.A. PlanApo objective lens and were collected by using a Roper Cascade 512B camera. All cells were imaged in DB buffer (10 mM Na_2HP0_4, 0.2 mM $CaCl_2$, 2 mM $MgSO_4$ [pH 6.5]) in Lab-Tek II chambers (Nalge Nunc International).

Supplemental Data

Supplemental Data including figures and movies referred to in the text are available at http://www.developmentalcell.com/cgi/content/full/8/4/467/DC1.

Acknowledgments

The authors wish to thank Dr. D. Robinson for providing the GFP-myosin II expression vector and for helpful discussions and critical reading of the manuscript. They also thank Dr. R. Firtel for providing the *pi3k*$^{1\text{-}2-}$ cells. This work was supported by National Institutes of Health grants GM28007, GM34933, and GM071920 to P.D. and Department of Defense grant DAM 17-03-1-0195 to F.V. Sequence data for *D. discoideum* was obtained from the dictyBase website at http://www.dictybase.org/.

Received: July 16, 2004
Revised: January 12, 2005
Accepted: February 22, 2005
Published: April 4, 2005

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